RGB-D–Guided Bowl Grasping and Placement

with a UR5e

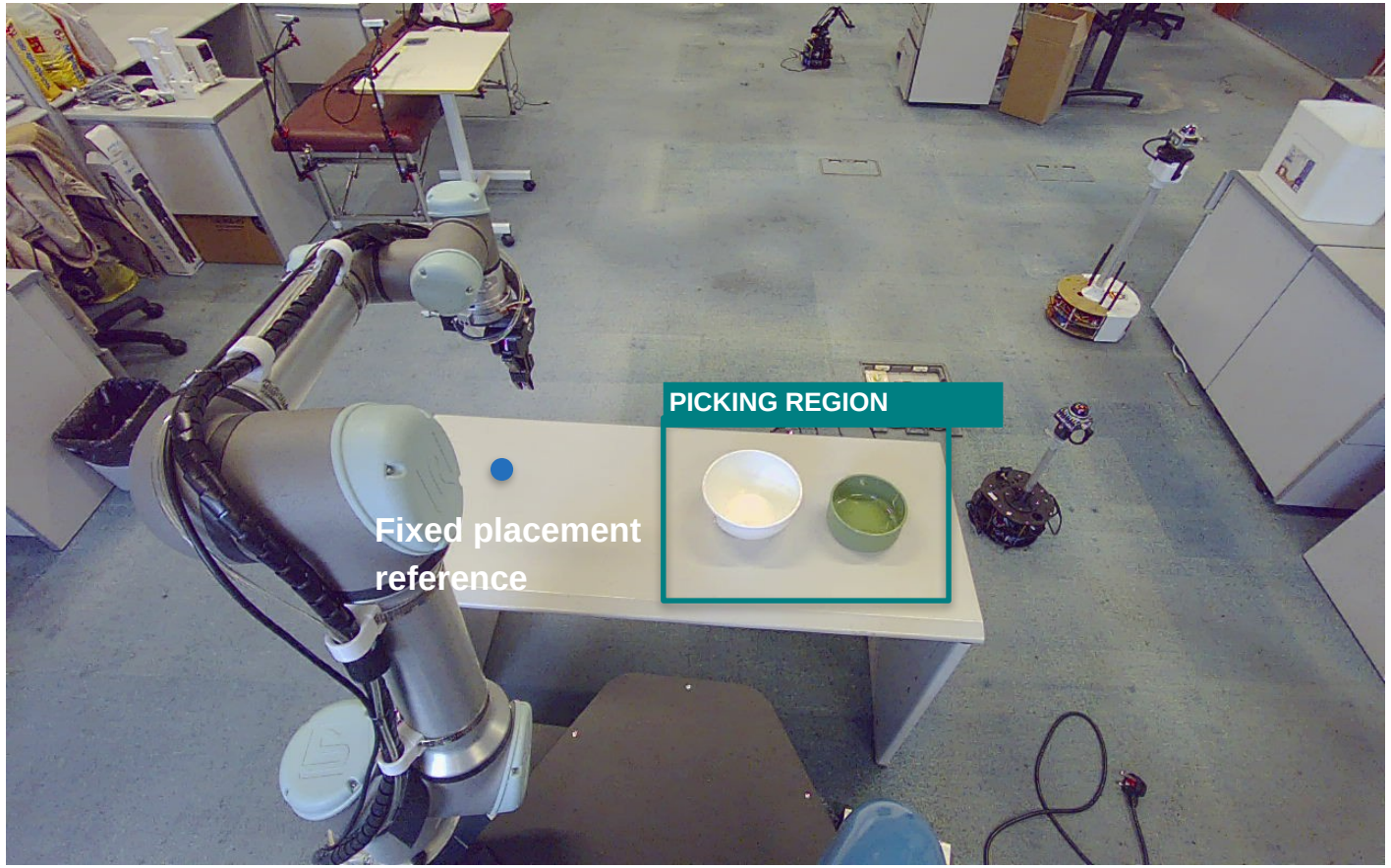
From single contacts to candidate-based planning and repeated observation

TWO BOWLS · ONE ROBOT · VERIFIED EXECUTION



A defined tabletop task, with separate pick and place regions

UR5e + parallel gripper + Orbbec Gemini2L + ROS 2 Humble



1 Detect

Observe the bowls inside the picking region.

2 Grasp and place

Use a rim contact and a stored placement pose.

3 Observe again

Acquire a new scene after release and retreat.

What changed from the initial grasping pipeline

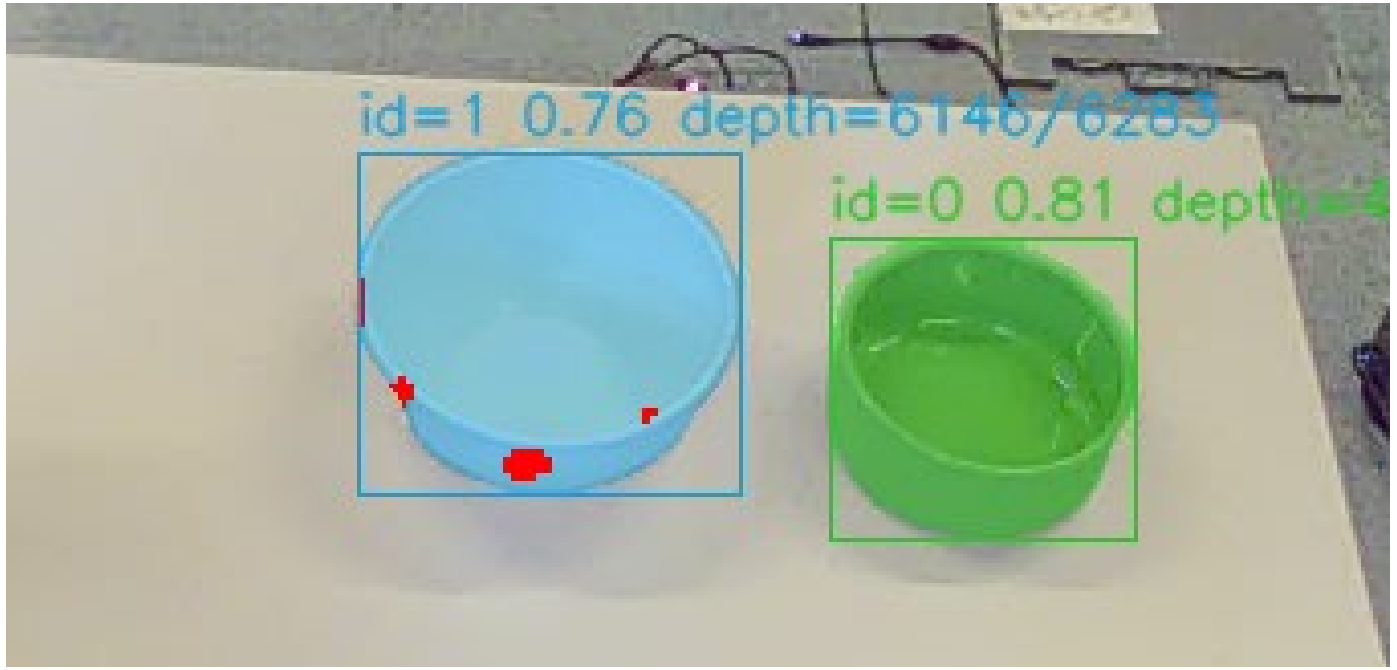
Three ROS packages retained; perception, geometry and execution extended.

Area	Earlier capability	Current implementation
Perception	Single-target output	Multiple instances + depth diagnostics
Rim geometry	One preferred contact	Up to 8 supported candidates per bowl
Motion	Approach / descend / lift	Six-stage grasp, transfer and placement
Task execution	Reuse one target snapshot	One bowl per round; then observe again
Operation	Separate manual processes	Mode isolation, logs and process supervision

Contribution: an integrated, testable manipulation pipeline with explicit failure handling.

Two bowls are segmented and reconstructed independently

One synchronized RGB-D observation | 27 Sep 2026 | plan-only dataset



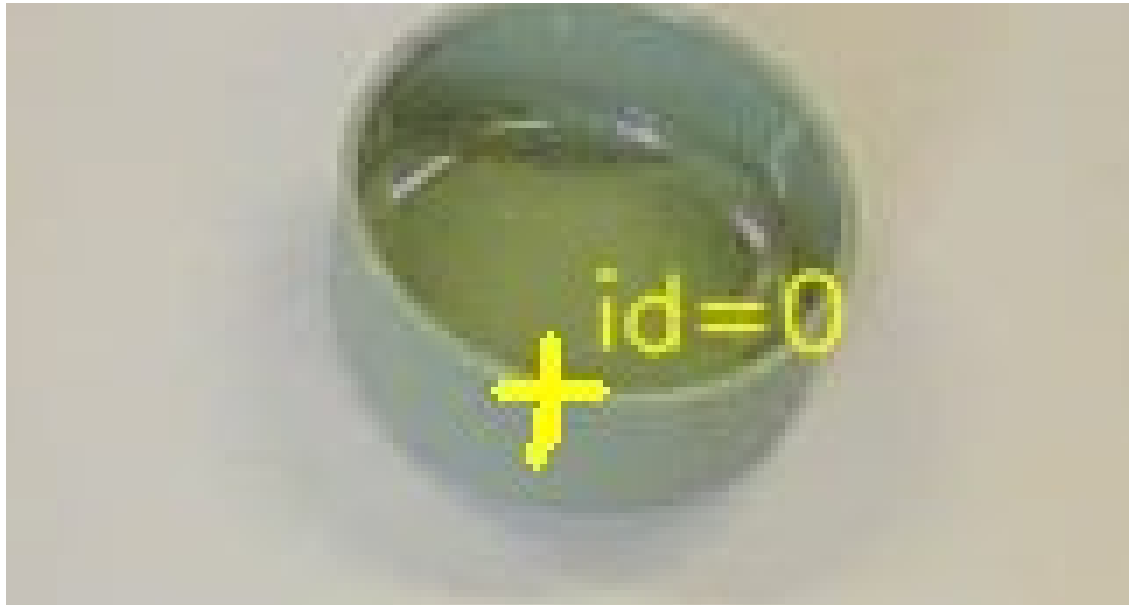
ORIGINAL SEGMENTATION OVERLAY · DISPLAY CROP

Recorded value	White bowl	Green bowl
Instance ID	1	0
Confidence	0.762	0.805
Valid depth points	6,146	4,595
Valid depth ratio	97.8%	100.0%

Detection, valid depth and a usable grasp are three different checks.

White and green bowls offer different wall evidence

Recorded geometry example | 23 Sep 2026, frame 01





Green bowl

30 wall points | 11.73° local wall tilt
0.609 mm plane residual

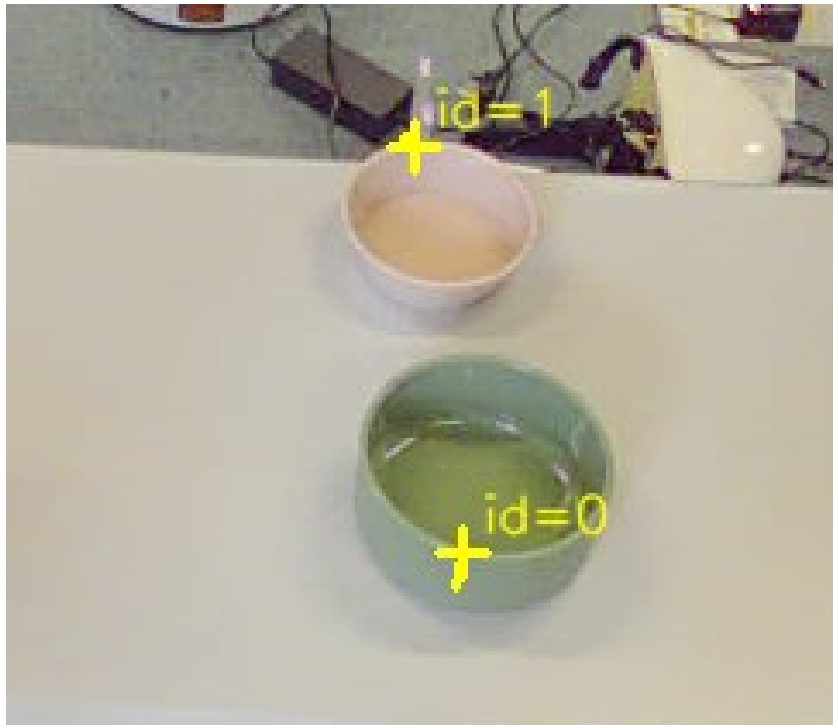
White bowl

33 wall points | 31.89° local wall tilt
0.549 mm plane residual

Measured wall tilt is not commanded tool tilt; the current tool remains vertical.

Both bowls were localized in three recorded observations

Recorded RGB-D replay | same geometric acceptance rules



MEASURED CONTACTS, FRAME 01

Frame	Green: points / tilt	White: points / tilt	Localized
01	30 / 11.73°	33 / 31.89°	2 / 2
02	42 / 13.82°	43 / 36.80°	2 / 2
03	50 / 8.82°	23 / 37.37°	2 / 2

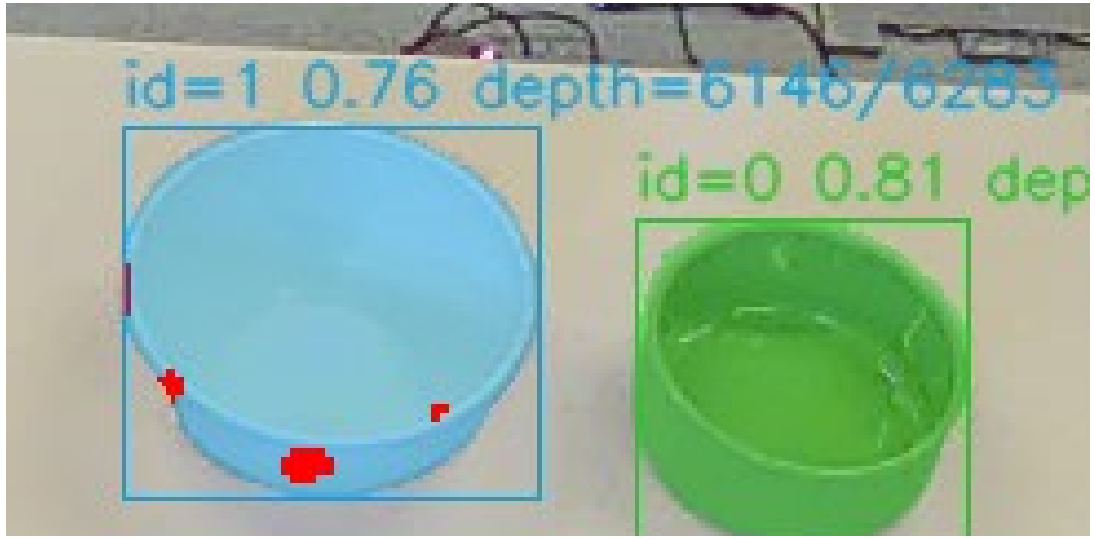
Quality gates

≥20 wall points · ≥6 mm wall span
≤2.5 mm residual · ≤12° normal change

Observed fitting residual: 0.46–0.74 mm. This is not ground-truth positioning accuracy.

Each bowl has its own candidates and processing order

27 Sep plan-only snapshot | complete six-stage plans found for both bowls



Bowl	ID	Candidates	XY distance	Plan
White bowl	1	8	0.654 m	6 / 6
Green bowl	0	8	0.827 m	6 / 6

Ordering uses geometry, not color or instance ID.

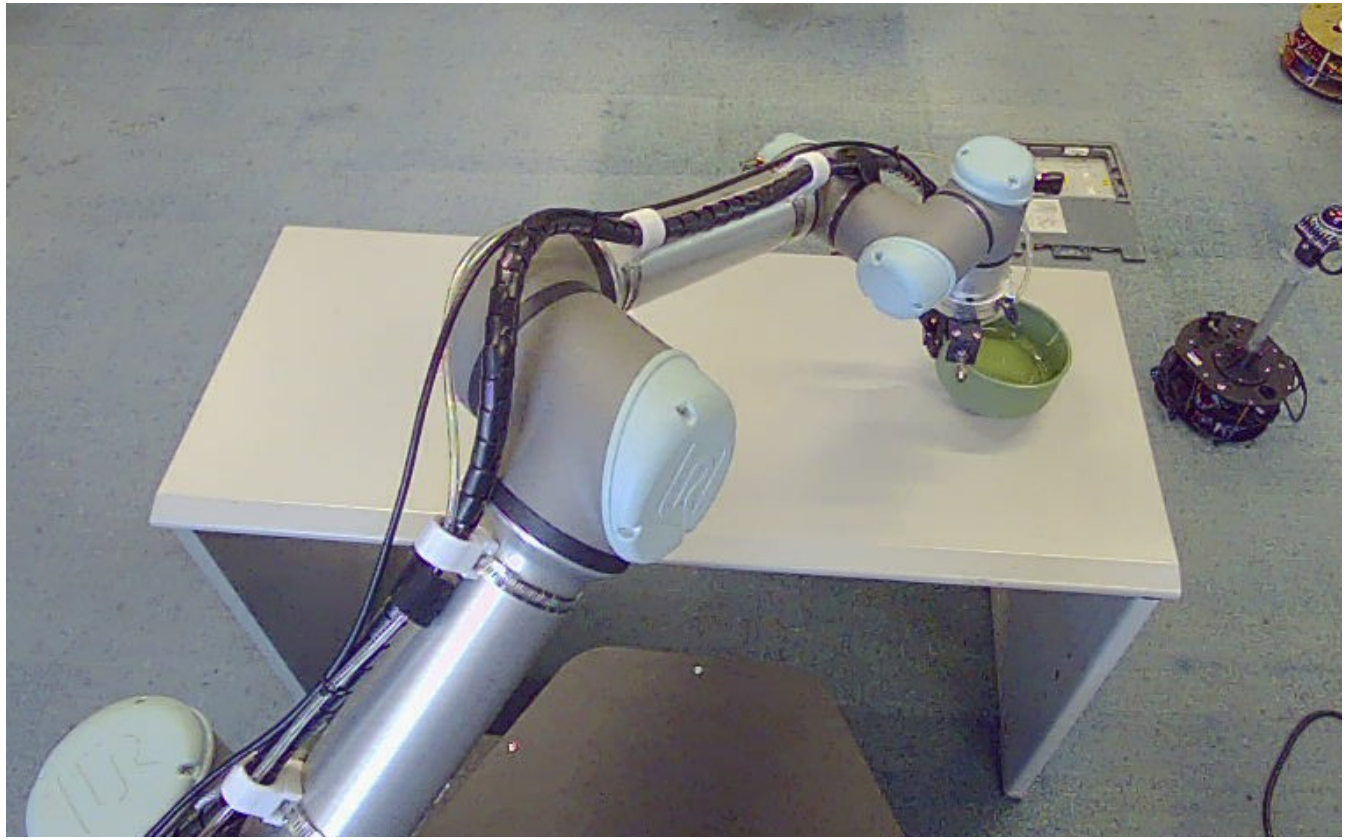
Before the first motion

Reject infeasible candidates. Execute only after the entire six-stage path passes.

Both targets planned successfully in this snapshot; planning success is not physical grasp success.

Feasible contacts still require feasible robot motion

Bounded IK search + OMPL transfer + Cartesian vertical segments



Joint-space consistency

Equivalent angles and URDF soft limits prevent avoidable joint winding.

Whole-path feasibility

Validate approach, descent, lift, transfer, lowering and retreat.

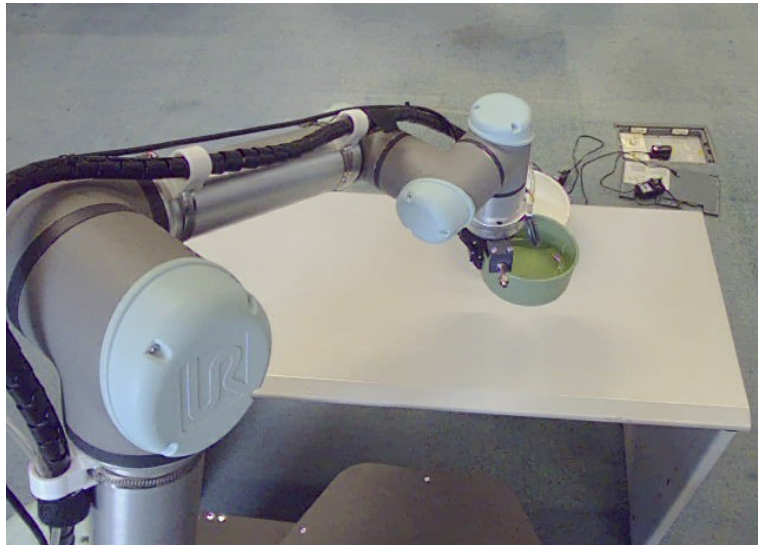
Bounded fallback

Try another candidate only before motion has started.

Candidate fallback is verified in isolated MoveIt tests; live fallback remains untested.

A stored TCP reference makes placement repeatable

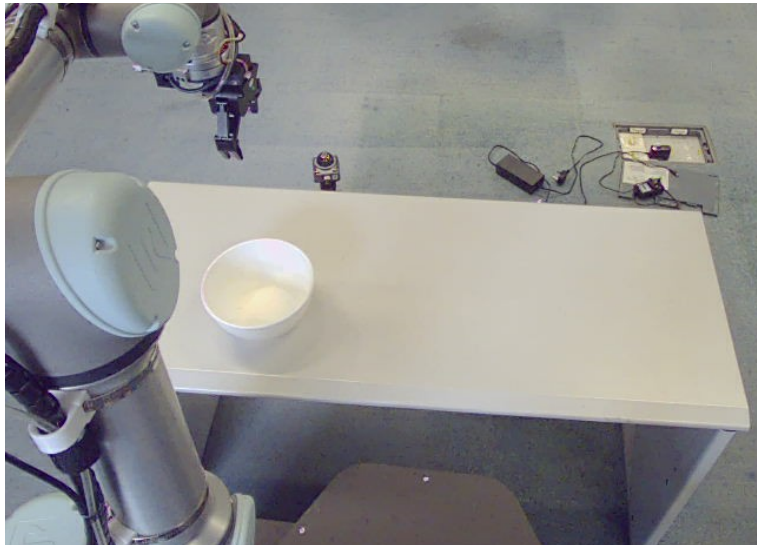
Fixed 6D pose | static TCP-to-grasp-center transform | MoveIt execution



LIFT · 23 SEP



PLACE · 23 SEP



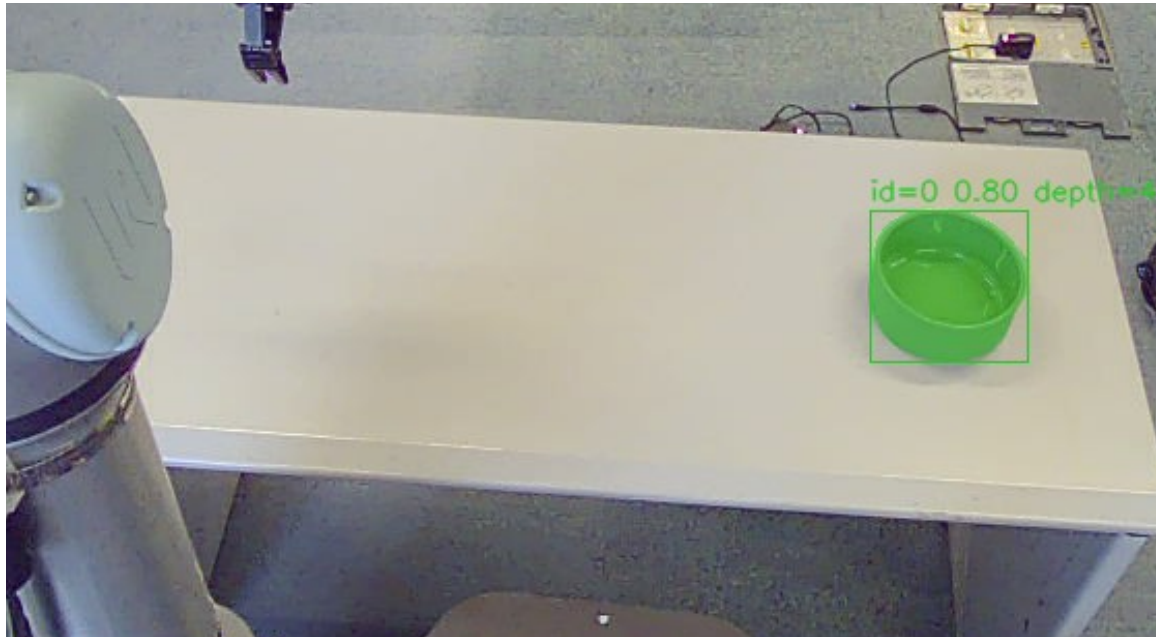
RETREAT · 23 SEP

01 Approach **02 Descend** **03 Lift** **04 Transfer** **05 Lower** **06 Retreat**

10 cm approach / lift | 20 cm placement descent | 1 s hold before release

One completed bowl, then a new observation

Post-fix green-bowl run | 27 Sep 2026



ROUND 1 · TARGET DETECTED

1 bowl completed

6 motion stages executed



ROUND 2 · EMPTY PICKING REGION

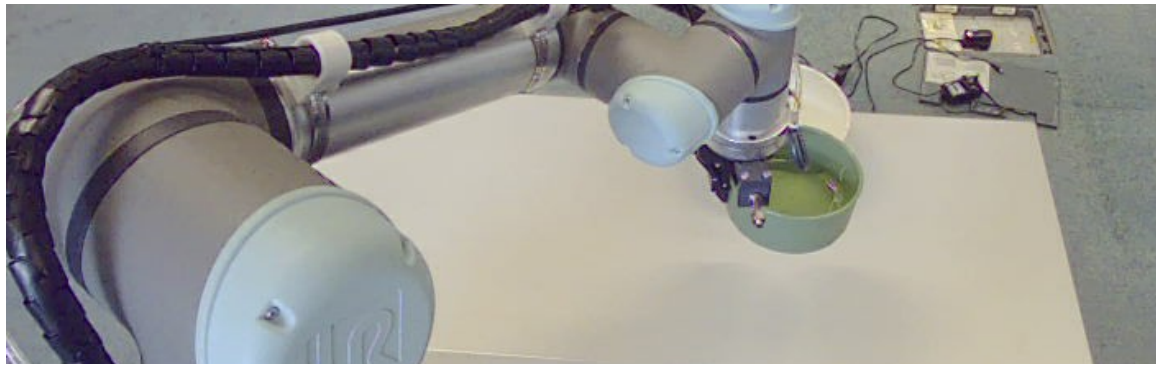
Fresh capture after retreat

0 targets reported by the next observation

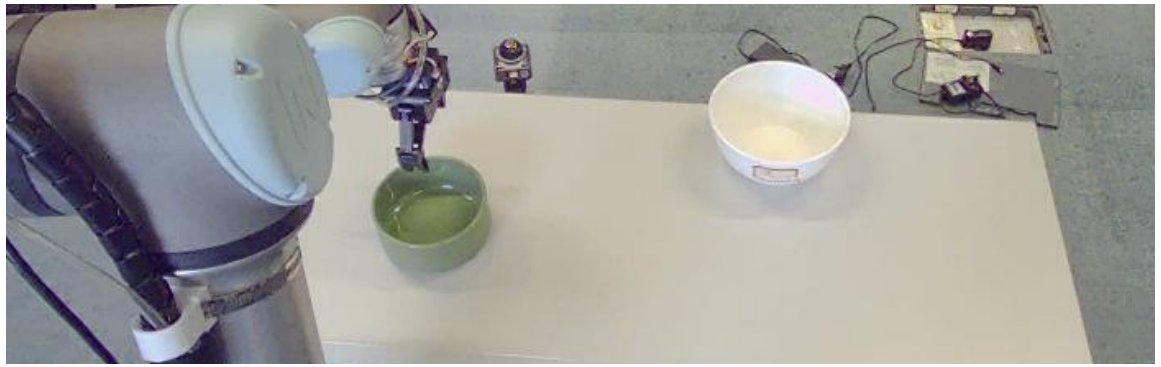
New timestamps, new candidates and a measured robot start state for every round.

Both bowl types were lifted and placed in supervised trials

23 Sep 2026 | observations from separate execution and recovery steps



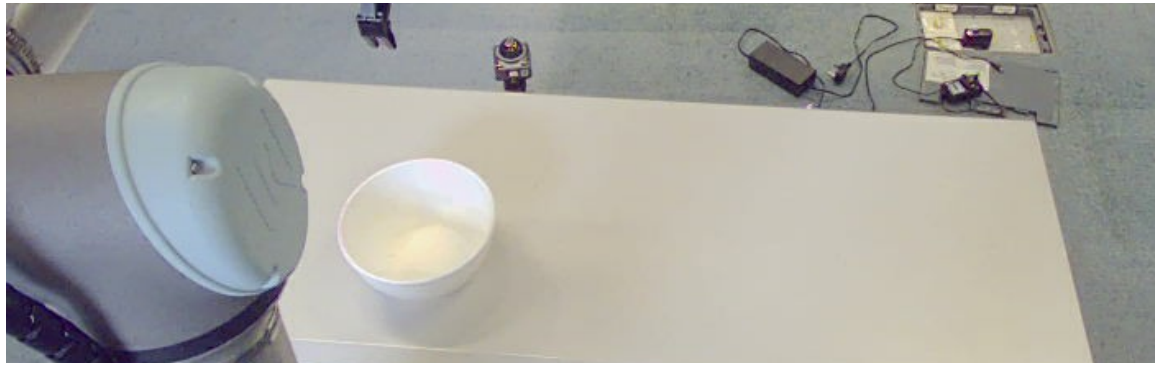
GREEN · LIFTED



GREEN · PLACED



WHITE · LIFTED



WHITE · PLACED

Human intervention occurred between these steps; this is not an uninterrupted two-bowl cycle.

The fixed placement point exposed a multi-bowl limitation

A successful first placement can occupy the destination needed by the next bowl.



What happened

The placed green bowl occupied the fixed destination.

What the trial required

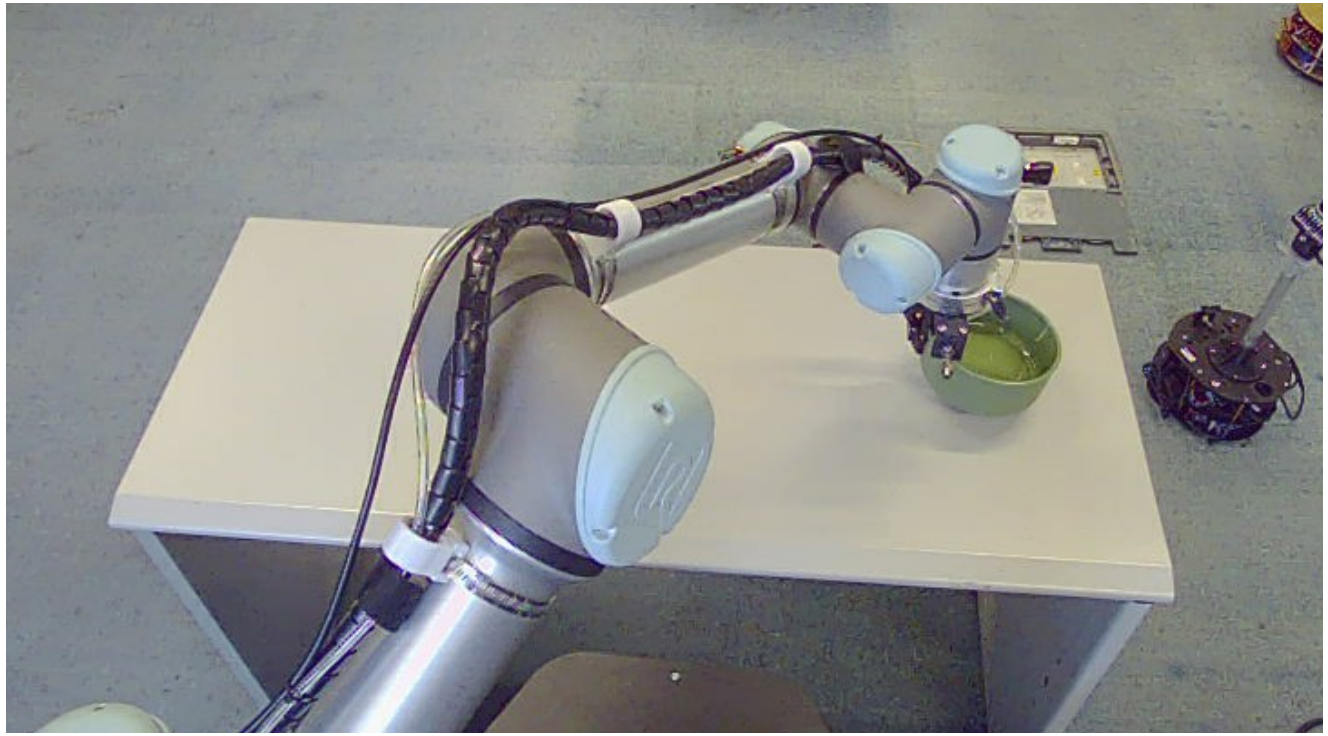
An operator cleared the destination before the white-bowl trial.

Not yet implemented:
placement occupancy sensing
or automatic slot allocation

Separating the picking ROI prevents re-picking; it does not prove placement capacity.

Post-fix execution completed a full cycle and re-observation

Selected run: execute_02_after_fix | real robot and camera



GREEN-BOWL MANIPULATION · ORIGINAL CAMERA IMAGE

Recorded field	Result
Outcome	completed
Completed cycles	1
Observation rounds	2
Candidate attempts	1
Cleanup errors	None reported

Command completion is logged; independent grasp / placement sensing is not available.

An interrupted trial led to stronger process supervision

The first 27 Sep white-bowl run was not counted as a complete success.



WHITE BOWL LIFTED · LATER RETREAT WAS CANCELLED

Observed gap

The runner exited while the independent executor remained active.

Implemented response

Process-identity monitoring

Explicit termination-signal handling

Task-owned process cleanup

After motion starts: stop and report errors; do not automatically release, switch candidates or regrasp.

Software validation is distinct from physical success

Recorded test results | suite counts are not combined into a grasp success rate

Validation layer	Result	Date	What it verifies
Regression tests	503 passed	27 Sep	Logic, state transitions and failure handling
Isolated ROS startup	7 passed	27 Sep	Controller checks and startup behavior
Launch contracts	32 checks	27 Sep	Parameters, namespaces and supervision
MoveIt integration	15 checks	25 Sep	Real planning with simulated execution
Physical trials	Images + logs	23 / 27 Sep	Supervised lifting, placement and retreat

Physical success rate remains unmeasured; automated test coverage is a separate result.

What is demonstrated, and what needs the next experiment

A working manipulation system with explicit evidence boundaries

Demonstrated

Two-instance perception and local geometry
Candidate-based complete-path planning
Physical lifts and placements
Fresh observation after a completed cycle

Next controlled comparison

Single vs. multiple candidates
Frozen targets vs. fresh observations
Repeated white / green bowl trials
Placement occupancy and grasp sensing

Measure: physical grasp success · full-cycle success · cycle time · human interventions

Goal: quantify the benefit of the added functions, not only demonstrate that a run can finish.

Appendix A — Current configuration and interfaces

Values describe the current project configuration, not universal operating limits.

Item	Current setting	Meaning
Candidate interface	/rim_locator/grasp_candidates	Versioned JSON grouped by object
Candidates per bowl	Up to 8; 4 mm separation	Bounded local contact alternatives
Tool / planning frame	grasp_center / tool0	Connected by fixed TF
Approach / lift	0.10 m	Along base_link Z
Place descent / dwell	0.20 m / 1.0 s	Fixed reference, then release
Target age ceiling	600 s	New capture still required each round

No added table model or gripper envelope; URDF robot geometry and trajectory checks remain active.

Appendix B — Reproduction and source records

Root-level scripts and matching New_run wrappers use the same implementation.

OFFLINE BUILD AND VALIDATION

```
./run_pipeline_place_offline_test.sh
```

HARDWARE PREPARATION → PREVIEW → EXECUTION

```
./run_robot_prepare.sh
./run_pipeline_place_test.sh --vision-config <site_roi.json>
./run_pipeline_place_execute.sh --vision-config <site_roi.json>
```

Site ROI: src/yolo_vision/config/live_pick_roi_20260927.json

Primary evidence: docs/live_demo_verification_20260927.md
Recorded images and logs: debug_output/live_demo_20260927_gjgip1/

English slides · Chinese speaker notes · original experimental images · traceable source files